Arash Nabbi

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Education

PhD. Medical Science, Cancer Biology — September 2010-January 2017
University of Calgary, Cumming school of Medicine, Calgary, AB
Thesis: "The role of the ING3 epigenetic regulator in prostate cancer"

PharmD. Pharmacy — January 2004-December 2009
Azad University of pharmaceutical sciences, Tehran, Iran
Thesis: "The study of Noscapine-induced apoptosis in Lymphoblasts derived from patients with leukemia"

Research Experience

Post-doctoral Fellow. Princess Margaret Cancer Centre — From August 2022
University Health Network, Toronto, ON
Mentor: Dr. Trevor Pugh

Post-doctoral Fellow. Princess Margaret Cancer Centre — September 2017- August 2022
University Health Network, Toronto, ON
Mentor: Dr. Trevor Pugh

PhD Student. Arnie Charbonneau Cancer Institute — September 2010-January 2017
University of Calgary, Calgary, AB
Supervisor: Dr. Karl Riabowol

PharmD Student. Razi Institute of Drug Research — May 2007-December 2009
Iran University of Medical Sciences, Tehran, Iran
Supervisor: Drs. Massoud Mahmoudian and Nahid Rahbar

Teaching Experience

Graduate Teaching Assistant. — September 2015-April 2016
Human Integrative Physiology. Bachelor of health sciences.
University of Calgary, Calgary, AB

Graduate Teaching Assistant. — September 2014-April 2015
Human Integrative Physiology. Bachelor of health sciences.
University of Calgary, Calgary, AB

Clinical Experience

Clinical pharmacist. Shoppers drug mart — From May 2021
Toronto, ON

Pharmacy Intern. Sandstone enhanced care solutions — January 2017-August 2017
Calgary, AB

Pharmacist in-charge. Mansoor Pharmacy — September 2008-January 2010
Tehran, Iran

Publications

Manuscripts in revision/preparation

1. **Nabbi, A.,** Beck, P., Delaidelli, A., Oldridge, D. A., Sudhaman, S., Zhu, K., Yang, S. Y. C., Mulder, D. T., Bruce, J. P., Paulson, J. N., Raman, P., Zhu, Y., Resnick, A. C., Sorensen, P. H., Sill, M., Brabetz, S., Lambo, S., Malkin, D., Johann, P. D., Kool, M., Jones, D. T. W., Pfister, S. M., Jäger, N. & Pugh, T. J. Transcriptional immunogenomic analysis reveals distinct immunological clusters in pediatric nervous system tumours. *In review. Nature Cancer.*
2. **Nabbi A**, Danesh A, Espin-Garcia O, Pedersen S, Wellum J, Fu LH, Geoerger B, Marshall LV, Trippett T, Rossato G, Pugh TJ, Hutchinson KE. Multimodal immunogenomic biomarker analysis of tumours from pediatric patients enrolled to a phase 1-2 study of single-agent atezolizumab. *In review. Nature Cancer.*

Peer reviewed papers (*: co-first author)

3. **Nabbi, A.*,** Beck, P.*, Delaidelli, A., Oldridge, D. A., Sudhaman, S., Zhu, K., Yang, S. Y. C., Mulder, D. T., Bruce, J. P., Paulson, J. N., Raman, P., Zhu, Y., Resnick, A. C., Sorensen, P. H., Sill, M., Brabetz, S., Lambo, S., Malkin, D., Johann, P. D., Kool, M., Jones, D. T. W., Pfister, S. M., Jäger, N. & Pugh, T. J. Transcriptional immunogenomic analysis reveals distinct immunological clusters in pediatric nervous system tumours. 2022.09.20.508719 Preprint at https://doi.org/10.1101/2022.09.20.508719 (2022)
4. Beck, P., Selle, B., Madenach, L., Jones, D. T. W., Vokuhl, C., Gopisetty, A., **Nabbi, A**., Brecht, I. B., Ebinger, M., Wegert, J., Graf, N., Gessler, M., Pfister, S. M. & Jäger, N. The genomic landscape of pediatric renal cell carcinomas. iScience 25, (2022).
5. Das A, Sudhaman S, Morgenstern D, Coblentz A, Chung J, Stone SC, Alsafwani N, Liu ZA, Abu Al Karsaneh O, Soleimani S, Ladany H, Chen D, Zatzman M, Cabric V, Nobre L, Bianchi V, Edwards M, Sambira Nahum LC, Ercan AB, **Nabbi A**, et al. Genomic predictors of response to PD-1 inhibition in children with germline DNA replication repair deficiency. Nat Med. 2022;28:125–35.
6. Fink D, Yau T, **Nabbi A**, Wagner B, Wagner C, Hu SM, et al. Loss of Ing3 Expression Results in Growth Retardation and Embryonic Death. Cancers (Basel). 2019 Dec 29;12(1).
7. Dantas A, Al Shueili B, Yang Y, **Nabbi A**, Fink D, Riabowol K. Biological Functions of the ING Proteins. Cancers (Basel). 2019 Nov 19;11(11).
8. McClurg UL, Chit NCTH, Azizyan M, Edwards J, **Nabbi A**, Riabowol KT, Nakjang S, McCracken SR, Robson CN. Molecular mechanism of the TP53-MDM2-AR-AKT signalling network regulation by USP12. Oncogene. 2018;37(34):4679–91.
9. McClurg UL*, **Nabbi A***, Ricordel C*, Korolchuk S, McCracken S, Heer R, Wilson L, Butler LM, Irving-Hooper BK, Pedeux R, Robson CN, Riabowol KT, Binda O. Human ex vivo prostate tissue model system identifies ING3 as an oncoprotein. Br J Cancer. 2018 06;118(5):713–26.
10. Wang F, Wang AY, Chesnelong C, Yang Y, **Nabbi A**, Thalappilly S, Alekseev V, Riabowol K. ING5 activity in self-renewal of glioblastoma stem cells via calcium and follicle stimulating hormone pathways. Oncogene. 2018 Jan 18;37(3):286–301.
11. **Nabbi A**, McClurg UL, Thalappilly S, Almami A, Mobahat M, Bismar TA, Binda O, Riabowol KT. ING3 promotes prostate cancer growth by activating the androgen receptor. BMC Med. 2017 16;15(1):103.
12. Almami A, Hegazy SA, **Nabbi A**, Alshalalfa M, Salman A, Abou-Ouf H, Riabowol K, Bismar TA. ING3 is associated with increased cell invasion and lethal outcome in ERG-negative prostate cancer patients. Tumour Biol. 2016 Jul;37(7):9731–8.
13. McClurg UL, Harle VJ, **Nabbi A**, Batalha-Pereira A, Walker S, Coffey K, Gaughan L, McCracken SRC, Robson CN. Ubiquitin-specific protease 12 interacting partners Uaf-1 and WDR20 are potential therapeutic targets in prostate cancer. Oncotarget. 2015 Nov 10;6(35):37724–36.
14. Thakur S, **Nabbi A**, Klimowicz A, Riabowol K. Stromal ING1 expression induces a secretory phenotype and correlates with breast cancer patient survival. Mol Cancer. 2015 Aug 27;14:164.
15. **Nabbi A**, Almami A, Thakur S, Suzuki K, Boland D, Bismar TA, Riabowol K. ING3 protein expression profiling in normal human tissues suggest its role in cellular growth and self-renewal. Eur J Cell Biol. 2015 May;94(5):214–22.

Review articles

1. Jones DTW, Banito A, Grünewald TGP, Haber M, Jäger N, Kool M, Milde T, Molenaar JJ, **Nabbi A**, Pugh TJ, Schleiermacher G, Smith MA, Westermann F, Pfister SM. Molecular characteristics and therapeutic vulnerabilities across paediatric solid tumours. Nat Rev Cancer. 2019 Aug;19(8):420–38.
2. **Nabbi A**, Satpathy Sh, Riabowol K. Demethylating agents as epigenetic anticancer therapeutics. Curr Can Ther Rev. 2013; 9: 24-33.
3. Satpathy S, **Nabbi A**, Riabowol K. RegulatING chromatin regulators: post-translational modification of the ING family of epigenetic regulators. Biochem J. 2013 Mar 15;450(3):433–42.

Book chapters

1. **Nabbi A**, Riabowol K. Isolation of Nuclei. Cold Spring Harb Protoc. 2015 Aug 3; 2015(8):731–4.
2. **Nabbi A**, Riabowol K. Isolation of Pure Nuclei Using a Sucrose Method. Cold Spring Harb Protoc. 2015 Aug 3;2015(8):773–6.
3. **Nabbi A**, Riabowol K. Rapid Isolation of Nuclei from Cells In Vitro. Cold Spring Harb Protoc. 2015 Aug 3;2015(8):769–72.

Knowledge Dissemination

Invited talks and lectures

1. **Transcriptional analysis reveals distinct microenvironmental subgroups across pediatric nervous system tumors.**
   The 20th International Symposium on Pediatric Neuro-Oncology. June 2022
   Hamburg, Germany (Invited short talk)
2. **The immunogenomic landscape of pediatric primary solid tumors.** March 2021
   Roche IMCORE pediatric workshop.
   Virtual. (Invited short talk)
3. **The immunogenomic landscape of pediatric primary solid tumors.** September 2021
   2019 AACR Advances in Pediatric Cancer Research.
   Montreal, Quebec. (Invited short talk)
4. **Immunogenomic characterization of pediatric tumor microenvironment**. December 2018
   Hopp Children's Cancer Centre (KiTZ).
   Heidelberg, Germany (Invited speaker)
5. **ING3: A novel co-activator of Androgen Receptor.** May 2017
   Pharmaceutical outcomes program, BC Children's hospital research institute.
   Vancouver, BC. (Invited speaker)
6. **Diuretics and the kidney**. March 2016
   MDSC404 course. Calgary, AB. (University lecture)
7. **Nonrespiratory functions of the lungs**. January 2016
   MDSC404 course. Calgary, AB. (University lecture)
8. **Biomarker discovery: Lessons learned from prostate cancer study**. May 2015
   Lung Cancer research group meeting. Calgary, AB. (Invited speaker)

Poster presentations

1. **Landscape of infiltrating immune repertoire in pediatric solid tumors.**
   2019 American Association of Cancer Research annual meeting. Atlanta, Georgia
2. **Immunogenomic characterization of pediatric solid tumors.**
   2018 Terry Fox Research Institute Ontario Node Research Symposium. Dec 2018. Toronto. ON
3. **The ING3 epigenetic regulator is a novel prognostic biomarker for primary prostate cancer.** 2015
   Canadian cancer research conference. Nov 2015. Montreal, QC.
4. **INhibitor of Growth 3 (ING3) modulates Androgen Receptor activity in prostate cancer**.
   SACRI Research Day. January 2015, Calgary, AB.
5. **The effects of INhibitor of Growth | 3 (ING3) on prostate cancer cell survival and invasion**.
   EACR23, Jul 2014, Munich, Germany.
6. **Deciphering the role of ING3 on androgen receptor activity in prostate cancer cells**.
   2013 Alberta Cancer Foundation Conference, Banff, AB.
7. **Noscapine induces apoptosis on lymphoblasts derived from childhood acute lymphoblastic leukemia patients**.
   7th International Symposium on Targeted Anticancer Therapies 2009, Amsterdam, Netherlands.

Awards and recognitions

AACR Scholar-in-Training Award (250 USD). September 2019
Advances in Pediatric Cancer Research, Montreal, QC
Princess Margaret Post-Doctoral Fellowship award (50,000 CAD). January 2018
Princess Margaret Cancer Centre, Toronto, ON
Queen Elizabeth II scholarship (15,000 CAD). September 2016
Faculty of Graduate Studies, University of Calgary, Calgary, AB
Queen Elizabeth II scholarship (15,000 CAD). September 2015
Faculty of Graduate Studies, University of Calgary, Calgary, AB
MDSC Graduate Program travel award (1,500 CAD) June 2014
University of Calgary, Calgary, AB
CIHR Training Program Travel Award-ACHRI (750 CAD) June 2014
University of Calgary, Calgary, AB
Program Specific Award (PSA) For first author publication (500 CAD) January 2013
Medical Science Graduate Program, University of Calgary, Calgary, AB

Program Specific Award (PSA) (2,550 CAD) January 2012
Medical Science Graduate Program, University of Calgary, Calgary, AB

Professional Certificates

Licensed under Ontario College of Pharmacists (#625769) March 2018
Injectable Medication and Vaccine Administration Training Program February 2017
Standard First Aid - CPR/AED Level C March 2021
Pharmacy Examination Board of Canada certificate (#42298) December 2016

Active Memberships

Society for Immunotherapy of Cancer (SITC)
Ontario College of Pharmacists. Member (#625769)
American Association of Cancer Research (AACR)

Skills

Technical skills
DNA library preparation, targeted capture sequencing, antibody validation, tissue culture, western blotting, qPCR, immunoprecipitation, immunofluorescence, molecular cloning, lentiviral preparation
Computer skills
R, Python, Bash, Adobe Illustrator, Jupyter notebooks, AWS, Docker
Language skills
English, German, Farsi

September 30, 2022